

Product Requirements Document (PRD)
Arogya – AI-Powered Holistic Health Super App

Version Information

Field	Value
Product Name	Arogya
Version	1.0
Document Type	Product Requirements Document (PRD)
Platform	Web Application (Responsive)
Frontend	React.js
Backend	Supabase
Database	PostgreSQL (Supabase)
Authentication	Supabase Auth
Deployment	Vercel
Repository	GitHub

1. Executive Summary

Vision

Create India's most intelligent and emotionally engaging health super app that unifies physical health, mental wellness, women's healthcare, beauty, AI companionship, accessibility, and telehealth into one seamless ecosystem.

Unlike traditional health applications that solve only one problem, **Arogya** becomes a lifelong AI companion that motivates users to build healthier habits through personalization, gamification, and predictive insights.

2. Problem Statement

Today's healthcare experience is fragmented.

Users need multiple apps for:

- Nutrition
- Fitness
- Sleep

- Mental Health
- Women's Health
- Beauty Care
- Doctor Consultation
- Habit Tracking

Switching between apps creates:

- Poor engagement
 - Lost health data
 - No personalized journey
 - Lack of motivation
 - Inconsistent routines
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3. Product Goals

Primary Goals

- Create one unified health platform
- Improve preventive healthcare
- Increase user engagement using gamification
- Build emotional connection using AI
- Support accessibility for every user

Business Goals

- Increase Daily Active Users
 - Increase retention
 - Increase habit completion
 - Enable future telehealth marketplace
 - Build scalable healthcare ecosystem
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4. Target Audience

Primary Users

Age:

18–45

Includes:

- Students
 - Working Professionals
 - Fitness Enthusiasts
 - Women
 - Families
 - Elderly Users
-

Secondary Users

- Doctors
 - Nutritionists
 - Fitness Trainers
 - Therapists
 - Caregivers
-

5. Product Pillars

Pillar 1 — Core Health Foundation

Purpose:

Build the essential healthcare system.

Modules

Nutrition Architect

Features

- Calorie Calculator
- BMI Calculator
- Water Intake
- Meal Planner
- Macro Tracker
- Nutrition Dashboard

Algorithms

- BMR
- TDEE

- Calorie Deficit
 - Macro Distribution
-

Fitness Flow

Features

- Workout Plans
- Daily Exercises
- Step Counter
- Progress Charts
- Goal Tracking

Algorithms

- Exercise Recommendation
 - Calories Burned
 - Progress Prediction
-

Preventive Lens

Features

- Risk Assessment
- Health Score
- Lifestyle Evaluation

Algorithms

- Risk Scoring
 - Health Index
-

Vitals Sync

Features

- Blood Pressure
- Heart Rate
- Blood Sugar
- Oxygen Level
- Weight History

Future

Wearable Integration

Women's Health

Modules

- Period Tracker
- Ovulation Prediction
- Pregnancy Tracker
- Hormonal Tracker

Algorithms

- Cycle Prediction
 - Fertility Window
 - Pregnancy Timeline
-

Pillar 2 — Mind & Lifestyle

Purpose

Improve emotional and lifestyle wellness.

Mood Care Sync

Features

- Daily Mood Check
- Emotion Journal
- Mood Analytics

Algorithms

- Mood Trend Detection
 - Emotional Pattern Recognition
-

Stress Radar

Features

- Stress Questionnaire
- Stress Score

- Recommendations
-

Sleep Sanctuary

Features

- Sleep Logging
- Sleep Quality Score
- Bedtime Reminder

Algorithms

- Sleep Efficiency
 - Sleep Debt
-

Beauty & Wellness

Modules

- Skin Glow Tracker
- Hair Health
- Body Care
- Wellness Rituals

Recommendations

- Hydration
 - Routine Builder
 - Product Suggestions
-

Pillar 3 — Engagement & AI

Purpose

Increase retention.

Gamification

Features

Daily

- Missions
- Streaks

- XP
- Coins
- Levels

Weekly

- Challenges
- Leaderboard

Rewards

- Badges
- Achievements

Community

Features

- Support Groups
- Discussion
- Challenges

AI Health Coach

Features

- Personalized Advice
- Daily Motivation
- Voice Interaction
- Health Diary
- Habit Reminder

Future

LLM Integration

Pillar 4 — Accessibility & Expansion

Purpose

Universal healthcare accessibility.

Accessibility

Features

- Voice Navigation
 - Large Text
 - Screen Reader Friendly
 - Emotional Audio
-

Telehealth

Features

- Doctor Directory
 - Appointment Booking
 - Chat
 - Video Consultation
-

Health Galaxy

Future Feature

Interactive 3D visualization of

- Health Progress
 - Achievements
 - Wellness Journey
-

6. Functional Requirements

Authentication

- Email Login
 - Google Login
 - Password Reset
 - User Profiles
-

Dashboard

Displays

- Health Score
- Today's Tasks

- Water Intake
 - Calories
 - Mood
 - Sleep
 - Streak
-

Notifications

- Water Reminder
 - Exercise Reminder
 - Medicine Reminder
 - Sleep Reminder
 - Period Reminder
-

Reports

Weekly

Monthly

Yearly

Export PDF

7. Non-Functional Requirements

Performance

- Page Load < 2 seconds
- API Response < 300 ms

Availability

99.9%

Security

- HTTPS
- JWT Authentication
- Row Level Security (Supabase)
- Encrypted Passwords

Scalability

Support

100,000+ Users

Accessibility

WCAG Compliance

Responsive

Desktop

Tablet

Mobile

8. Database Design (Supabase)

Tables

Users

Profiles

Health Metrics

Meals

Workouts

Mood Logs

Sleep Logs

Women's Health

Badges

Achievements

Challenges

Notifications

Appointments

Doctors

Community Posts

AI Conversations

9. Algorithms

Health Score

Based on

- BMI
- Sleep
- Mood
- Exercise
- Water
- Nutrition

Calorie Recommendation

Uses

- BMR
- TDEE

Workout Recommendation

Input

- Goal
- BMI
- Age
- Experience

Output

Personalized workout

Mood Prediction

Based on

Previous mood history

Sleep

Stress

Activity

Cycle Prediction

Average cycle length

Period history

Ovulation prediction

Gamification

XP Calculation

Daily Streak

Badge Unlock

Leaderboard Ranking

10. Tech Stack

Frontend

- React.js
- React Router
- Context API / Redux
- Tailwind CSS
- Chart.js
- Framer Motion

Backend

- Supabase
- PostgreSQL
- Edge Functions

Authentication

- Supabase Auth

Deployment

- Vercel

Version Control

- Git
 - GitHub
-

11. User Journey

Landing Page



Signup/Login



Complete Health Profile



Dashboard



Choose Goals



Daily Health Tracking



AI Recommendations



Earn XP



View Analytics



Improve Health Score

12. Success Metrics (KPIs)

- User Registration Rate
 - Daily Active Users (DAU)
 - Monthly Active Users (MAU)
 - 7-day and 30-day Retention
 - Average Session Duration
 - Daily Habit Completion Rate
 - Health Score Improvement
 - Challenge Participation Rate
 - AI Coach Engagement
 - User Satisfaction (NPS)
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13. MVP Scope (Version 1.0)

Must Have

- User Authentication
- Dashboard
- Nutrition Tracker
- Fitness Tracker
- Mood Tracker
- Sleep Tracker
- Women's Health
- Health Score
- AI Health Coach (basic)
- Gamification (streaks, badges)
- Reports
- Responsive UI

Nice to Have

- Community
- Voice Assistant
- Doctor Booking
- Notifications
- Device Sync

Future Releases

- Telemedicine
- Wearable Integration
- AI Image Analysis (skin/food)
- Health Galaxy (3D)
- Multi-language Support
- Family Health Profiles
- Insurance Integration

14. Risks & Mitigations

Risk	Impact	Mitigation
Inaccurate health recommendations	High	Use evidence-based formulas, clearly label as informational, recommend consulting professionals for medical decisions
Low user engagement	Medium	Gamification, AI reminders, personalized goals
Data privacy concerns	High	Supabase Row Level Security, encryption, secure authentication, minimal data collection
Scalability issues	Medium	Optimize queries, caching, serverless architecture on Vercel
Complex UI	Medium	Progressive onboarding, clean navigation, accessibility testing

15. Future Roadmap

Phase 1 (MVP): Authentication, Dashboard, Core Health, Mood & Sleep, Women's Health, Basic AI Coach, Gamification.

Phase 2: Community features, Voice Assistant, Push Notifications, Doctor Directory, Telehealth Booking.

Phase 3: Wearable integrations, AI-powered image analysis, predictive health insights, multilingual support.

Phase 4: Health Galaxy 3D experience, family health management, insurance and pharmacy integrations, advanced AI companion.

Deliverables

- **Frontend:** React.js application with responsive UI
- **Backend:** Supabase (Auth, PostgreSQL, Storage, Edge Functions)
- **Repository:** GitHub with version control
- **Deployment:** Vercel CI/CD
- **Documentation:** PRD, SRS, API documentation, Database schema, UI/UX prototype, Testing report, Deployment guide

This PRD provides a complete blueprint for developing **Arogya** from MVP through future releases while keeping the product scalable, user-centric, and suitable as a major academic or portfolio project.